Mahmudul Hasan Khan

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PROFESSIONAL SUMMARY

Heterogeneous catalysis researcher completing a Chemical Engineering PhD (expected May 2027), with four years of catalyst synthesis, fixed-bed reaction testing, and characterization focused on renewable feedstock hydroprocessing. Developed a kinetic method to quantify competitive adsorption in multicomponent feeds, diagnosed 30–70% activity loss from catalyst reduction and coking over 16 h on stream, and generated the 15+ catalyst dataset that identified support descriptors for tuning selectivity. Achieved 85% BTX selectivity at full conversion from a bio-derived feed at ambient pressure. Seeking early-career industrial R&D opportunities.

EDUCATION

Doctor of Philosophy, Chemical Engineering May 2027 (*Expected*)
Iowa State University (ISU), Ames, IA, USA | GPA: 3.96/4

Master of Engineering, Chemical Engineering May 2025
Iowa State University (ISU), Ames, IA, USA | GPA: 3.96/4

Bachelor of Science, Chemical Engineering February 2021
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh | GPA: 3.62/4

TECHNICAL SKILLS

Catalysis and Reaction Engineering: Reaction kinetics and mechanism development; fixed-bed reactor design and operation; catalyst deactivation diagnosis and stabilization strategies; transport analysis and verification of kinetic regime (internal and external mass transfer limitations); structure–activity–selectivity relationships.
Catalyst Synthesis: Incipient wetness impregnation; wet impregnation; solution-phase metal oxide reduction; supported metal oxide catalyst synthesis and screening; pretreatment protocols.
Characterization and Analysis: Gas chromatography (GC, GC-MS); chemisorption (TPR, TPD, pulse titration); BET physisorption; X-ray diffraction (XRD); X-ray photoelectron spectroscopy (XPS); Raman spectroscopy; FTIR; TEM; SEM; ICP-OES.
Data and Statistics: MATLAB; Python (NumPy, SciPy, Pandas); JMP; R; MS Excel; OriginPro; Power BI; CasaXPS; Design of Experiments (DoE: full/fractional factorial); Statistical Process Control (SPC); root cause analysis.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant August 2022–Present
Department of Chemical and Biological Engineering (CBE), ISU, Ames, IA, USA

- Competitive Adsorption of Lignin-derived Oxygenates over Molybdenum Trioxide (MoO_3) Catalyst during Hydrodeoxygenation
  - Designed and ran 100+ single- and co-feed HDO experiments (e.g., acetone + anisole) over bulk MoO_3 at 330 °C and 1 atm to understand competition effects inaccessible to single-compound studies.
  - Developed a kinetic framework for extracting relative adsorption constants, quantifying 6× higher adsorption propensity for acetone on the catalyst surface compared to anisole, evidence that single-feed kinetics can misrepresent catalyst performance on real mixed feeds.
- Deactivation of MoO_3 Catalyst during HDO of Lignin-derived Oxygenates
  - Diagnosed the root cause of MoO_3 deactivation during lignin-oxygenate HDO; quantified $30–70\%$ activity loss over 16 h on stream and linked it to catalyst reduction and coking through spent-catalyst characterization.
  - Designed and evaluated 6+ pretreatment strategies (redox and carburization), showing none stabilized the catalyst and ruling out pretreatment as a mitigation path for this system.
- Identification of Sabatier Descriptors for HDO Activity and Selectivity on Supported Metal Oxide Catalysts
  - Synthesized and characterized 15+ supported MoO_3 catalysts and benchmarked them for carbonyl HDO, generating the dataset that established support electronegativity and point of zero charge as tunable selectivity descriptors.

- Ambient-Pressure Conversion of Bio-derived 3-methylanisole to Benzene, Toluene, and Xylenes (BTX) over MoO_3 Catalysts
  - Used design of experiments (DoE) to screen 10+ MoO_3-based catalysts for ambient-pressure conversion of 3-methylanisole, a sugar fermentation byproduct, to BTX aromatics with 85% selectivity at full conversion for the lead formulation.
  - Developed an integrated process concept linking 3-methylanisole conversion with waste-biomass utilization to enable production of value-added BTX aromatics from renewable feedstocks.

Teaching Assistant January 2023–December 2023

Department of Chemical and Biological Engineering, ISU, Ames, IA, USA

- CH E 4060: Environmental Chemodynamics – Led weekly help sessions for a class of 30 students, mentoring them through problem-solving approaches; prepared homework solution sets and graded assignments with feedback.
- CH E 4260: Chemical Engineering Laboratory II – Led distillation and absorption unit operations labs for 24 senior students; supervised equipment operation, enforced laboratory safety procedures, and graded technical reports.

HSE Intern October 2021–February 2022

Chevron, Dhaka, Bangladesh

- Reviewed internal safety audits and recommended corrective actions, contributing to improved compliance with Chevron's safety standards and reduced flagged items in annual review.
- Updated the regulatory compliance workbook to reflect the latest government policies and developed concise safety briefing materials used in monthly "Safety Moment" meetings, enhancing workforce engagement on operational risk awareness.

Research Assistant March 2021–September 2021

Institute of Water and Flood Management (IWFM), BUET, Dhaka, Bangladesh

- Water Quality Monitoring (WQM) and Modeling of Transboundary Rivers in Bangladesh
  - Identified pollution trends and seasonal variations across two major transboundary river basins by assessing 100+ water and 20+ soil samples; co-authored a policy-oriented technical report and proposed a streamlined WQM framework for adoption at the national level.

SELECTED PUBLICATIONS AND PRESENTATIONS

- **Khan**, M.H., Fard, Z.H., Kohler, A.J., Roling, L.T. and Shanks, B.H., 2025. Elucidating the Competitive Hydrodeoxygenation of Lignin-Derived Oxygenates over Bulk MoO_3 Catalyst through Kinetic Analysis. *ACS Catalysis*, 15(17), 15488–15501.
- Kohler, A.J., **Khan**, M.H. and Shanks, B.H., 2025. Identification of Sabatier Descriptors for Hydrodeoxygenation Activity and Selectivity on Supported Molybdenum Oxide Catalysts. *ACS Catalysis*, 15(5), 4317–4330.
- **Oral presentation: Khan**, M.H., Kohler, A.J., and Shanks, B.H. 2024. Kinetic Analysis of Competitive Hydrodeoxygenation of Lignin-derived Molecules over Bulk MoO_3 Catalyst. American Institute of Chemical Engineers (AIChE) Annual Meeting.
- **Poster presentation: Khan**, M.H. and Shanks, B.H. 2025. Elucidating Molybdenum Trioxide (MoO_3) Deactivation during the Ambient Pressure Hydrogenation of Lignin-Derived Model Compounds. The 29th North American Catalysis Society Meeting.

GRANT PROPOSALS, AWARDS AND FELLOWSHIPS

- Second Place, CBE Perfect Pitch Award, CBE, ISU May 2026
- Lead author, USDA-NIFA Sun Grant Program (North Central Region) proposal; PI: Prof. Brent Shanks; Requested: $60,000; Status: Not selected for funding January 2026
- George W. Parrott Centennial Graduate Fellowship, CBE, ISU August 2022
- CBE Excellence Fellowship, CBE, ISU August 2022

LEADERSHIP AND PROFESSIONAL MEMBERSHIPS

- **Leadership:** Vice President (2024–25) and Professional Development Chair (2023–24), Chemical Engineering Graduate Student Organization, ISU.
- **Memberships:** NSF Engineering Research Center for Biorenewable Chemicals (CBiRC), ISU; American Institute of Chemical Engineers (AIChE); Great Plains Catalysis Society (GPCS); Tau Beta Pi, Iowa Alpha.